

AHT 212 – AFRICAN ARCHAEOLOGY / HUMAN EVOLUTION

Detailed Revision Notes in Point Form

Prepared from the uploaded AHT 212 module. The notes retain the module's organization, terminology and major examples rather than replacing them with outside material.

COURSE OVERVIEW

- The course analyzes African archaeology from the Cenozoic period to the origin and development of African art.
- It examines the development of modern humans and the forces or actions associated with human origins.
- It explains why Africa has played a major role in human modernity and the development of art.
- It considers how humans developed abilities to deal with their environment and how environmental conditions influenced human development.
- It investigates fossilization and the geological context required for interpreting fossils.

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1. FOSSILS IN GEOLOGICAL CONTEXT

Meaning and importance of fossils

- Fossils are preserved remnants of once-living organisms and provide evidence about past life.
- Palaeontology is the study of fossils and the information that can be extracted from them.
- A fossil without geological context has limited value because its age, environment and association with other organisms cannot be properly assessed.
- Geology helps establish preservation conditions, age and the environment in which fossils were deposited.

Taphonomy and fossilization

- Taphonomy studies what happens to an organism from death until its discovery.
- Death may result from disease, injury, predation and other causes.
- Predators may leave marks such as bite marks on bones.
- After death, decomposition begins and microbes, bacteria and insects accelerate the process.
- Scavengers such as hyenas may consume soft tissues and may also chew bones.
- Durable tissues such as teeth and dense bones are more likely to survive.
- Erosion and trampling can also destroy even durable bones.
- Burial is essential for preservation: remains may be covered by sand, silt, mud, gravel, dust or volcanic ash.
- Burial commonly occurs in floodplains, lake shores and swamps where sedimentation occurs.
- After burial, minerals from soil or groundwater may replace the original inorganic tissues, producing petrification.
- Teeth and bones are commonly preserved; collagen, the organic component, normally degrades.
- Exceptional preservation may include skin or hair, frozen animals such as mammoths, mummified people, and coprolites (fossilized feces).

Why context matters

- Context establishes the position of a fossil within rock layers and its relationship to other sites.
- Context allows archaeologists and geologists to interpret age, environment and associations.

Stratigraphy

- Stratigraphy is the study of the distribution and sequence of layers in soil or rock.
- It helps archaeologists determine where a find occurs within the sequence of deposits.
- Uniformitarianism proposes that processes operating today can help explain geological and fossil processes of the past.
- Original horizontality: sediments are originally laid down approximately parallel to the earth's surface.
- Later deformation can result from events such as earthquakes and volcanic eruptions.
- Superposition: when undisturbed, older layers occur below younger layers.
- Cross-cutting relationships: a feature that cuts another feature is younger than the feature it cuts.
- Faunal succession: fossil animal communities occur in recognizable sequences through strata.
- Index fossils are characteristic fossils used to identify or correlate particular layers.
- Comparison of stratigraphic sequences between sites helps establish relative age relationships.

2. GEOLOGIC TIME SCALE (GTS)

- The geologic time scale provides a chronological framework linking stratigraphy to time.
- It provides an ordered sequence of geological intervals and the history of life on Earth.
- It contains both relative time divisions and numerical/absolute age estimates.
- The major nested divisions are eons, eras, periods and epochs.
- The Earth is presented in the module as approximately 4.5 billion years old.
- Human and primate evolution is emphasized within the Cenozoic, beginning about 65 million years ago.

Major divisions emphasized in the module

- Precambrian Eon: approximately 4.5 billion to 543 million years ago; includes Hadean, Archaean and Proterozoic.
- Phanerozoic Eon: approximately 543 million years ago to the present; includes Paleozoic, Mesozoic and Cenozoic eras.
- Cenozoic: approximately 65 million years ago to the present; the module divides it into Tertiary and Quaternary periods.
- Tertiary epochs: Paleocene, Eocene, Oligocene, Miocene and Pliocene.
- Quaternary epochs: Pleistocene and Holocene.
- The module gives the Pleistocene as about 1.8 million years to 10,000 years ago and the Holocene as 10,000 years ago to the present.

3. EARTH IN THE CENOZOIC

Continents and land masses

- Continents have shifted through geological time and did not always occupy their present positions.
- Around 200 million years ago, major land masses included Laurasia and Gondwanaland.
- Laurasia included present-day North America and Asia; Gondwanaland included Africa and South America.
- By about 50 million years ago, Africa and South America had separated and continental positions were changing toward the modern arrangement.
- Africa became connected to Asia through the Near East.
- Glacial events between about 15 million years ago and 10,000 years ago periodically lowered sea levels and could expose additional land or create land bridges.

4. PALAEO-ENVIRONMENTAL RECONSTRUCTION

Environmental proxies

- Past environments are reconstructed using geological and biological evidence known as environmental proxies.
- Proxies can be used to reconstruct temperature, sea level, plant communities and animal communities.
- Evidence from different sites can be combined through time and across space to infer past environmental conditions.

Oxygen isotopes, temperature and sea level

- Oxygen isotope curves use the ratio of stable oxygen isotopes, especially ^{16}O and ^{18}O , as a climate proxy.
- ^{18}O is heavier than ^{16}O .
- During cold periods, more ^{16}O becomes locked in ice, leaving relatively more ^{18}O in seawater.
- Higher $^{18}\text{O}/^{16}\text{O}$ ratios therefore indicate colder conditions and lower sea levels in the module's interpretation.
- Lower ratios indicate warmer conditions and higher sea levels.

- Marine shells can preserve oxygen isotope signals that are measured through time.
- Lake sediments can also be used to reconstruct local climates; the module gives Lake Victoria as an East African example.
- The Plio-Pleistocene is described as having repeated shifts between glacial and interglacial conditions.
- Odd-numbered oxygen isotope stages represent interglacial periods; even-numbered stages represent glacial periods.
- Oxygen isotope curves help correlate broad events, including human migrations, while regional microclimates may require finer-scale evidence.

Paleosols and loess

- Paleosols are ancient soils whose structure reflects environmental conditions at the time of formation.
- They can indicate conditions experienced by ancient plants and herbivores.
- Paleosols can help correlate strata between archaeological sites.
- Loess consists of windblown sediment and can contribute to stratigraphy and paleoclimate reconstruction.

Vegetation evidence

- Plant microfossils, root casts, plant imprints and pollen can preserve evidence of past vegetation.
- Bog, marsh and peat environments may preserve plant remains.
- Pollen can indicate the plants growing in and around a site, although wind can transport pollen from surrounding areas.
- Phytoliths are opaline silica bodies produced by grasses; their shapes can be characteristic of particular plants.
- Phytoliths can be recovered through fine, wet screening of sediments.
- Vegetation evidence has been used to investigate hominid and primate diets and the use of stone tools.

Stable carbon isotope ratios

- Stable carbon isotopes help reconstruct vegetation and environmental conditions.
- Plants use different photosynthetic pathways, including C3, C4 and CAM pathways.
- C3 plants include trees and shrubs and tend to have lower $^{13}\text{C}/^{12}\text{C}$ ratios.
- C4 plants include most tropical grasses and have relatively more ^{13}C .
- CAM plants include plants such as cactus and are adapted to conditions where reducing water loss is important.
- Carbon isotope ratios in fossil animal teeth can indicate the vegetation consumed by animals.
- Carbon isotope evidence from animal taxa and paleosols can help reconstruct vegetation maps.
- The distinction between C3 and C4 vegetation is especially useful for identifying open savanna versus more wooded environments.

Animal communities

- Some animals have strong habitat preferences that can be used to infer past environments.
- Hippos and crocodiles indicate association with water; monkeys are associated with wooded environments.
- Animals adapted to open terrain tend to have longer, more slender limbs, while forest-adapted animals tend to have shorter limbs.
- The animal-community approach considers all fauna from a site and time period rather than relying on one species.
- Relative abundance of different animals can provide a stronger environmental signal.
- Taphonomic processes must be considered because animals or their remains can be transported by water or other animals.

- Combined animal evidence can indicate habitats and available foods such as grasses and leaves.

Paleoethnobotanical evidence and charcoal

- Plant remains are useful for reconstructing past environments when compared with modern habitat preferences and geographical ranges.
- Depositional and destructive processes can distort the original plant community.
- Charcoal is burnt organic material commonly derived from wood or grain.
- Sources of charcoal include burning vegetation, firewood, burnt structural timber and accidentally charred grain.
- Charcoal can be dated by radiocarbon (carbon-14) methods.
- Charcoal can also help identify the plant or tree species used.

5. THE EARLIEST APES

- The living ape genera listed in the module are Hylobates (gibbons and siamangs), Pongo (orangutan), Gorilla, and Pan (bonobos and common chimpanzees).
- The fossil record of hominoids shows diverse adaptive radiations.
- Fossil apes appeared during the Early Miocene, approximately 23–16 million years ago.
- Early Miocene fossil hominoids were predominantly represented in African sites including Kenya, Uganda, Tanzania and Namibia.
- The module describes early Miocene Africa as having extensive forest and moist woodland.
- Dental apes had apelike teeth but monkey-like postcranial skeletons.
- They were relatively small-bodied compared with modern apes and lacked the suspensory shoulder associated with brachiation.
- They walked in a plantigrade fashion, on the soles of the feet.
- Proconsul is identified as a major example, living in Africa about 18–20 million years ago.

6. THE BASIS FOR HUMAN EVOLUTION – BIPEDALISM

Meaning of bipedalism

- Bipedalism is locomotion on two legs.
- Humans use bipedalism as their normal form of locomotion.
- Chimpanzees and gorillas can show facultative bipedalism, meaning they adopt two-legged locomotion temporarily for particular activities.
- Human striding gait involves alternating swing and stance phases.
- During swing, the leg moves forward; during stance, the leg supports the body and contributes to forward movement.
- Human gait includes heel strike and toe-off, features that distinguish it from the constantly flexed lower-limb posture of chimpanzees.

Theories for the origin of bipedalism

- Locomotor efficiency – Peter Rodman and Henry McHenry examined the energetic costs of bipedalism and quadrupedalism.
- Thermoregulation – Peter Wheeler proposed that an upright posture reduced the body area exposed directly to sunlight and increased exposure to cooling winds.
- Freeing the hands – upright locomotion allowed early hominins to carry food, resources and other objects.
- Postural feeding – Kevin Hunt proposed that bipedal posture could improve fruit gathering from small trees and increase foraging efficiency.

7. DIVERGENCE OF THE HUMAN LINEAGE

- Fossils such as Nakalipithecus from Kenya and Ouranopithecus from Greece are discussed as possible forms close to the last common ancestor of gorillas, chimpanzees and humans.
- The module presents molecular evidence for divergence of gorillas and chimpanzees from the lineage leading to humans between about 8 and 4 million years ago.
- The fossil record of gorillas and chimpanzees is limited partly because acidic rainforest soils can dissolve bones and because of sampling bias.
- As environments became drier, hominins adapted to environments outside the equatorial belt.

- Sahelanthropus tchadensis (about 7 million years ago) and Orrorin tugenensis (about 6 million years ago) are listed among the earliest hominins.

8. AUSTRALOPITHECINES

- The module places the origin of the hominid family between roughly 10 and 5 million years ago.
- The evolutionary history is presented as a branching 'bush' in which many species arose, some became extinct and others persisted.
- By about 2 million years ago, several hominid species coexisted in Africa.
- The module distinguishes large-brained Homo from australopithecines.
- Australopithecines are described as bipedal apes with modified dentition.
- They lived in relatively open environments including bushland and woodland savanna.
- Their teeth and jaws show adaptations requiring considerable grinding.
- The module describes australopithecines as social and principally vegetarian, while also noting debate about meat consumption.
- Male australopithecines are described as larger than females.
- Two broad forms are emphasized: gracile (slender) and robust.

Gracile versus robust australopithecines

- Gracile forms were smaller and more delicately built.
- Robust forms were larger and more massive, especially in the chewing apparatus.
- Robust species had larger grinding teeth, stronger jaws and more developed chewing muscles.
- Australopithecus africanus is given as a South African gracile species.
- Australopithecus robustus is given as a South African robust species.
- Australopithecus boisei is identified as an East African robust australopithecine.
- Both forms had brain sizes close to about 500 cm³ in the module.
- Robust forms had very large, flat molars and relatively small incisors and canines.
- The robust lower jaw reflects powerful chewing.
- A sagittal crest provided attachment for the temporal muscle in robust forms.
- Large temporal and masseter muscles contributed to enlarged cheekbones and a characteristic dished facial appearance.
- The module notes debate over diet, including evidence that robust australopithecines consumed tougher foods and challenges suggesting significant meat consumption.

Hands, locomotion and possible tool use

- The postcranial skeletons of gracile and robust forms were broadly similar.
- Australopithecine pelvises show adaptations for bipedalism.
- Some skeletal evidence also suggests climbing ability.
- Australopithecus afarensis had apelike hand features such as curved phalanges and a short thumb.
- Later robust forms had more humanlike hand features, including a longer, more mobile thumb and broader fingertips.
- The module notes debate over whether robust australopithecines could make stone tools.
- Simple bone tools such as digging sticks have been recovered in contexts associated with robust australopithecines, but their makers are uncertain.

9. ORIGINS OF THE GENUS HOMO

- The evolution of Homo is broadly divided into early, middle and late phases.
- Early Homo retained many similarities to early australopiths.
- Middle Homo developed anatomy more similar to modern humans but retained comparatively small brains.
- Late Homo developed larger and more complex brains, with language and culture becoming increasingly important.
- A major feature associated with Homo is increased cranial capacity.
- The module gives an average modern human cranial capacity of about 1,350 ml, with a broad range of roughly 1,000–2,000 cc.
- Possible Australopithecus ancestors are given as having approximately 350–500 ml brain sizes.
- Early Homo discoveries are associated with Olduvai Gorge.
- Distinctive Homo features include increased cranial vault, reduced lower facial prognathism, smaller premolars and molars, and changes in the molar row.

10. HOMO ERECTUS

- Homo erectus appears in Africa at around 1.8 million years ago in the module.
- It is presented as the first hominin species to leave Africa, around 1.7 million years ago.
- Important sites outside Africa include Dmanisi in Georgia and Sangiran in Indonesia.
- The species persisted for a very long period and appeared in Africa, Georgia, Southeast Asia and mainland Asia.
- Its long geographic and temporal range helps explain regional anatomical differences.

Anatomical features of Homo erectus

- The skull is described as robust, long, low and angular.
- A prominent supraorbital torus or brow ridge occurs above the eyes.
- Other thickened cranial structures include the angular torus and occipital torus.
- The forehead is low and sloping or receding.
- The rear of the skull has a distinctive shape with thickened sutural regions including sagittal and metopic keels.
- Brain size is given as approximately 700–1,200 cc, averaging about 900 cc.
- The jaw is robust and powerfully built.
- Premolars and molars have complex root systems.
- The incisors are described as concave with ridges producing a shovel-shaped form.

Body size and adaptation

- Complete H. erectus skeletons are relatively rare.
- KNM-WT 15,000, known as Turkana Boy or Nariokotome Boy, is highlighted as a major East African find.
- It was found west of Lake Turkana in 1984 by a team associated with Alan Walker and Richard Leakey.
- The module describes H. erectus as robustly proportioned, with some individuals quite tall.
- Long lower limbs and a narrow hip produced long, linear body proportions.
- These proportions are interpreted as adaptations to tropical environments and heat dissipation.
- The module suggests sweating and active daytime behavior as part of this adaptation.

Important Homo erectus sites

- Koobi Fora: associated with early *H. erectus* remains, including KNM-ER 3733.
- Nariokotome: associated with the Turkana Boy.
- Olduvai Gorge: OH 9 and OH 12 are highlighted for variation in cranial capacity.
- Olororgesailie: produced *H. erectus* remains dated in the module to about 900,000 years ago.
- Bouri Formation in Ethiopia and the Danakil Depression in Eritrea also produced *H. erectus* evidence.

11. ORIGINS OF MODERN HUMANS

- The module describes the origin of modern humans as a relatively recent evolutionary phenomenon.
- Anatomical, archaeological and genetic evidence is presented as pointing to Africa as the origin of Homo sapiens.
- The Out of Africa Model is presented as the current best explanation in the module.
- The model proposes a single African origin for Homo sapiens followed by migration into Eurasia.
- Evidence cited includes African fossils, African stone tools and artifacts, and DNA evidence indicating an African founding population.

12. MULTIREGIONAL CONTINUITY MODEL

- The model proposes that after Homo erectus left Africa and dispersed across the Old World, regional populations gradually evolved into modern humans.
- Some gene flow between geographically separated populations is proposed to have prevented complete speciation.
- All living humans are traced in the model to Homo erectus populations that left Africa nearly two million years ago.
- Natural selection acting in different regions is proposed to have produced regional variation.
- Modern Homo sapiens is therefore seen as emerging across the geographic range occupied by humans rather than in one locality.

13. OUT OF AFRICA MODEL

- Modern humans evolved relatively recently in Africa.
- Homo sapiens then migrated into Eurasia.
- Earlier populations descended from Homo erectus were replaced according to the model as presented in the module.
- Homo sapiens arose in one place, probably Africa, with the module geographically including the Middle East in this discussion.
- The model treats modern human variation as relatively recent.

14. EVIDENCE FOR THE ORIGIN OF MODERN HUMANS

Anatomical evidence

- Early hominins dispersed from Africa and developed regional morphological differences through genetic drift and natural selection.
- Asian populations are associated in the module with forms such as Peking Man and Java Man.
- European and western Asian populations developed into Neanderthals.
- Neanderthals had a long, low cranial vault, strong brow ridges and a projecting mid-face.
- They had a large nasal aperture, robust skeleton, thick limb bones, broad shoulders and relatively short forearms and lower legs.
- Early African Homo sapiens had a more vertical forehead, rounded occipital region and reduced brow ridge.
- They had a reduced facial skeleton, a chin and a less robust body plan.
- The module uses these differences to argue that Neanderthals and early modern humans followed separate evolutionary histories.

Archaeological evidence

- Middle Stone Age Homo sapiens in Africa and Middle Paleolithic Neanderthals in Europe had some similarities in archaeological assemblages.
- Their toolkits were dominated by flakes and showed limited variation over long periods.
- Bone, antler and ivory tools were uncommon in these assemblages.
- Burials generally lacked rich grave goods and obvious ritual indicators.
- Hunting was often focused on less dangerous species and fishing evidence was limited.
- Population densities were apparently low.
- Living structures were rare or absent and fireplaces were relatively rudimentary.
- Evidence for art and decoration was limited.

Behavioral modernity / Late Stone Age

- Around 40,000–50,000 years ago the archaeological record shows a major transformation in behavior and technology.
- There was increased technological diversity and innovation.
- Tools became more diverse and changed through time and space.
- Bone, antler and ivory were increasingly used alongside stone.
- Blade technology became important.
- Burials could contain ritual elements and grave goods.
- Living structures and better-designed fireplaces appeared.
- Hunting of dangerous animals and fishing occurred more regularly.
- Population densities increased.
- Art and personal adornment became widespread.
- Raw materials such as flint and shells were exchanged over distances.
- The module describes these developments as evidence of fully modern appearance and behavior.

15. EXPLANATIONS FOR THE EMERGENCE OF BEHAVIORAL MODERNITY

- Soffer's behavioral/social explanation emphasizes changes in social relations, including development of the nuclear family.
- Klein's biological explanation proposes that mutations and major neural reorganization improved information processing.
- The module notes that brains do not fossilize, making direct testing of neurological reorganization difficult.
- Archaeological evidence for ritual and art is used to infer advanced symbolic and language abilities.

16. GENETIC EVIDENCE

- Studies of genetic variation in modern humans are presented as supporting a recent origin of Homo sapiens consistent with Out of Africa.
- Mitochondrial DNA (mtDNA) is especially important in the module.
- Humans show relatively little genetic variation compared with chimpanzees.
- The highest level of mtDNA variation is reported in African populations.
- High African diversity is interpreted as evidence that Homo sapiens originated in Africa and had more time there to accumulate variation.
- The module reports estimates of an African origin between about 100,000 and 400,000 years ago based on genetic distance.
- It also discusses a relatively small founding population leaving Africa, with an estimate of 10,000–50,000 people between about 50,000 and 100,000 years ago.
- Neanderthal DNA was found to be substantially distinct from modern human DNA in the studies cited in the module.

Additional considerations

- Qafzeh and Skhul in Israel are discussed as sites with anatomically modern humans close to 100,000 years ago.
- Tabun Cave is discussed as having Neanderthal remains dated to about 110,000 years ago.
- Amud and Kebara are cited as evidence that Neanderthals and Homo sapiens overlapped in the Middle East.
- European evidence places fully modern humans in Europe around 35,000–40,000 years ago, associated with the Aurignacian tradition.
- The module discusses Châtelperronian assemblages in southern France as an example of debated interaction between Neanderthals and modern humans.
- The Lagar Velho specimen is discussed as a proposed Neanderthal/sapiens hybrid, although the module gives reasons for questioning that interpretation.
- The module notes that some Asian evidence has been used to support multiregional continuity, including proposed regional continuity in Java and possible mixed traits in Chinese fossils.

17. SUMMARY: ORIGIN OF MODERN HUMANS

- The majority of anatomical, archaeological and genetic evidence in the module supports the view that fully modern humans are a relatively recent evolutionary phenomenon.
- The Out of Africa Model is presented as the best explanation in the module.
- Africa is presented as the principal origin of Homo sapiens.
- Neurological, technological, cultural and symbolic innovations contributed to the success of fully modern humans.

18. BEHAVIOR AND EVOLUTION OF EARLY HOMININS

- Early hominins created and used tools.
- Chimpanzees also demonstrate tool use, so tool use is not uniquely human.
- Humans became unusually dependent on technology and its products.
- Stone tools are particularly important archaeologically because stone preserves better than materials such as wood, bark and leaves.
- Stone tools could be used for slicing, scraping and hammering and could help manufacture other implements.
- Stone-tool evidence provides important, but incomplete, information about subsistence activities.

19. ARCHAEOLOGY OF STONE TOOLS

- Early archaeological studies emphasized collecting and classifying stone tools by type.
- Contemporary approaches also examine tools in their subsistence and behavioral context.
- Questions include the breadth of diet, importance of particular subsistence activities, existence of home bases, exploitation of territory and range size.

20. GRAHAME CLARK'S FIVE MODES OF STONE TECHNOLOGY

- Mode 1: simple chopping tools produced by knocking a few flakes from a cobble.
- Mode 2: more extensive preparation and conceptualization, especially bifacial handaxes.
- Mode 3: prepared cores used to produce more standardized flakes that could be retouched into various artifacts.
- Mode 4: narrow blades struck from prepared cores.
- Mode 5: microlithic technology involving small, delicate artifacts.
- The modes describe technological characteristics of assemblages rather than rigid archaeological time periods.
- Earlier modes could persist after later modes appeared.
- Mode 1 appears in Africa around 2.6 million years ago and persisted for a very long time.
- Mode 4 blades were produced in Africa for more than 100,000 years but appear later in the European archaeological record.

21. STONE AGE TERMINOLOGY

- In sub-Saharan Africa, the period before agriculture and iron is referred to as the Stone Age.
- African Stone Age divisions emphasized in the module are Early Stone Age (ESA), Middle Stone Age (MSA) and Later Stone Age (LSA).
- In North Africa and Europe, comparable periods are often discussed as Lower Paleolithic, Middle Paleolithic and Upper Paleolithic.
- ESA begins with Mode 1 tools and includes the later appearance of Mode 2.
- The appearance of prepared-core technology (Mode 3) marks the transition toward the MSA in the module.
- Mode 4 is mostly associated with the Upper Paleolithic and LSA, although elements can occur in the MSA.

22. THE EARLIEST KNOWN TOOLS – OLDOWAN

- The earliest stone tools discussed are dated to approximately 2.6 million years ago.
- Important early locations include the Lower Omo Valley, Hadar and Gona in Ethiopia and the western shore of Lake Turkana.
- Some Lower Omo tools were made from small quartz pebbles fractured to create sharp edges.
- Many tools from about 2.6–1.5 million years ago were made from lava cobbles.
- The technology is called Oldowan after Olduvai Gorge in Tanzania.
- Olduvai examples from Bed I and lower Bed II are dated in the module to about 1.9–1.6 million years ago.

Oldowan artifact categories

- Tools: scrapers, choppers, discoids and polyhedrons.
- Utilized pieces: large flakes produced during manufacture that have useful sharp edges.
- Waste: small pieces produced during manufacture or retouching.
- Manuports: pieces of rock carried to a site without being modified.

Characteristics and uses of Oldowan technology

- Oldowan production was relatively opportunistic compared with later technologies.
- Tool forms flowed into one another and were produced for immediate or specific purposes.
- Flake production was a major goal; cores could be by-products of the flaking process.
- Variation in core and flake size partly reflects differences in available raw material.
- Flakes could be used for defleshing, shearing, cutting meat and modifying other tools.
- Hammerstones and heavy cores could be used to open bones for marrow or brain.
- Wood, bone and antler tools may also have existed, but preservation makes them difficult to identify.

Skill required in Oldowan flaking

- The core needed an acute edge, less than about 90 degrees, near the striking point.
- The hammer had to strike with a glancing blow near the edge.
- The blow needed to be directed through a high-mass area such as a ridge or bulge.
- Successful flaking therefore required knowledge of force, angle and the structure of the core.

Developed Oldowan

- Around 1.6 million years ago a Developed Oldowan industry appeared.
- It contained fewer choppers and more spheroids, subspheroids and small scrapers.
- Bifacial tools, including early handaxes and cleavers, appeared and later became characteristic of Acheulian technology.

Who made the tools?

- Homo and Australopithecus lived contemporaneously between about 2.6 and 1.5 million years ago.
- The available evidence in the module points toward Homo as the likely maker of the earliest stone tools, while not completely excluding other hominins.

23. ACHEULIAN TECHNOLOGY

- Acheulian is identified as Mode 2 technology.
- The earliest known collection in the module is from Konso-Gardula in Ethiopia, dated to about 1.4 million years ago.
- The name Acheulian comes from St. Acheul in northern France, where many handaxes were collected.
- The main innovation was the appearance of larger handaxes, picks and cleavers.
- Compared with Oldowan choppers, Acheulian handaxes required greater conceptualization and planning.
- Acheulian tools are generally associated with *Homo erectus*.
- The handaxe remained characteristic of Acheulian assemblages in Africa and Eurasia for a very long time.
- Manufacture became increasingly refined through time.
- Core preparation became increasingly important; the module links this development to the Levallois technique and later Mode 3 technologies.
- Acheulian sites in Africa listed include Olorgesailie, Muguruk, Kapthurin and Chesowanja.
- Outside Africa, Ubeidiya in Israel is highlighted as an early Acheulian site on a possible migration route.
- European examples include Isernia in Italy, Vertesszöllos in Hungary and Arago in France.

Functions of Acheulian tools

- The main hypothesis is use as axes or heavy-duty knives.
- They may have been used for slicing tough animal hides.
- Microwear studies indicate use on meat, bone, wood and hide.
- The combination of weight and sharp edges provided advantages over small flakes.

End of Acheulian

- The end of Acheulian industries around 300,000–200,000 years ago marked the transition from Early Stone Age/lower Paleolithic traditions to Middle Stone Age/Middle Paleolithic traditions.
- The MSA is presented as lasting roughly from 300,000 to about 40,000–50,000 years ago, with regional differences.

24. MIDDLE STONE AGE (MSA)

- The MSA is positioned between the ESA and LSA in African archaeology.
- It began around 300,000 years ago and ended around 50,000 years ago in the module, although regional dates vary.
- In Europe the broadly corresponding tradition is termed Middle Paleolithic.
- MSA assemblages include flakes and flake-blades, some retouched into scrapers, knives, points and backed pieces.
- The technology is associated with anatomically modern or nearly modern *Homo sapiens*.
- Early physical evidence is highlighted from Omo and Herto in Ethiopia.
- Modes 3 and 4 are important when discussing the archaeology of modern human origins.

Levallois technique

- Levallois is a prepared-core technique associated particularly with Mode 3.
- The core is prepared to create a controlled striking surface and convex form.
- Broad flakes can then be detached in a planned manner.

- Repeated removal around the core can produce multiple flakes of similar form.
- The technique is more economical in raw material use than earlier flaking methods.
- The resulting flakes can be retouched into a range of cutting, scraping and piercing implements.

MSA regional variation and innovation

- Regional styles include Still Bay and Howiesons Poort.
- The Middle Awash valley and Olorgesailie basin are described as important centers of behavioral innovation.
- Kapthurin, Twin Rivers and Kalambo Falls provide evidence of blades, backed pieces and other innovations.
- The module notes evidence of microlith production at Mumba in Tanzania around 65,000 years ago and in the Howiesons Poort industry at Border Cave.
- The MSA also shows increased diversity of food resources, including marine mammals, fish, shellfish and tortoises at some coastal sites.
- Chronological control is often difficult, especially at open-air sites, so detailed classification is approached cautiously.

25. LATER STONE AGE (LSA)

- The LSA is distinguished from the MSA by increased production of blades and microliths.
- Blades could be systematically removed from prepared cores.
- Microliths are small flaked tools and were often designed to be attached to wood or bone.
- Arrowheads are given as an example of microliths, appearing in the module around 13,000–10,000 years ago.
- Bone, ivory, antler and shell were increasingly used.
- These materials could be ground, polished and drilled.
- Examples include harpoons, awls, bone points and other specialized tools.
- Ostrich eggshell and bone beads indicate personal adornment.
- Bored stone digging-stick weights, tortoiseshell bowls and ostrich-eggshell containers are also highlighted.

26. COMPARING ESA, MSA AND LSA

- ESA: dominated by simpler Mode 1 and Mode 2 technologies, including Oldowan and Acheulian.
- MSA: associated strongly with prepared-core technologies such as Levallois, greater regional variation and increasing tool diversity.
- LSA: greater emphasis on blades, microliths, bone/ivory/antler/shell tools, personal adornment and specialized technologies.
- Technological change provides evidence for increasing planning, innovation and regional adaptation.
- The module emphasizes that earlier technologies could persist when they remained useful.

27. TECHNOLOGY AND HUMAN ADAPTATION

- Stone-tool invention and refinement helped hominins exploit different environments.
- Technology supported movement into new geographical areas.
- Changes in technology are associated with changes in brain size, cognition and modernity.
- Regional differences show that technology was shaped by environmental conditions, available resources and migration histories.
- Tool assemblages should therefore be studied not only as types but also in relation to subsistence, behavior and spatial organization.

28. AFRICAN PREHISTORIC ART

Meaning and scope

- African prehistoric art refers to Stone Age art in Africa.
- The module states that Africa has a very large number of prehistoric-art specimens.
- Art is presented as part of human lifestyle rather than simply decoration.
- Art includes painting, sculpture, drawing, pottery, bodily adornment, music, dance and other forms of creative expression.
- African prehistoric art includes rock images, ceramic forms, carved animal bones and other symbolic materials.
- The earliest artistic representations in the module include markings made with red ochre.

29. ORIGINS AND EARLY AFRICAN ART

- Pictographs and petroglyphs are described as important early forms of African art.
- Rock art has been found in areas including South Africa and Namibia.
- Images depict animals, people and human activities.
- The exact dating of the earliest African rock art is debated in the module, with suggested ages ranging from tens of thousands to much older evidence.
- Blombos Cave in South Africa is highlighted for engraved ochre pieces and shell beads.
- The module associates some Blombos evidence with approximately 70,000–75,000 years ago.
- Ochre was an important coloring material used in early artistic expression.

30. MATERIALS USED IN ROCK PAINTING

- Minerals from soil could be ground into powder and mixed with substances such as animal fats.
- Iron-containing minerals such as ochre produced various colors.
- Manganese and charcoal could produce black pigment.
- Vegetable dyes may also have contributed colors such as blue.

31. TECHNIQUES OF PREHISTORIC AFRICAN ART

Painting

- Durable paints were used on rock surfaces, cave walls and pottery.
- Crucibles and pestles found at settlements provide evidence connected with paint production.

Petrographs / petroglyphic engraving

- Engraving involves etching rock surfaces to create figures and other forms.
- Some engraved forms could become three-dimensional.
- The module describes early African petroglyphs as more than 6,000 years old.
- This is a graphical technique based on modifying rock surfaces.

Pottery and molding

- Pottery involved producing pots and other vessels from clay.
- The coil technique involved placing rows of clay on top of one another until the vessel was formed.
- Decoration could include burnishing, incision and comb-rolling while the clay remained workable.

- Human and animal figurines could also be molded from clay.
- A Nubian vessel dated in the module to 3800–3100 B.C. is cited as an example of decorated pottery.

Masks, figures and decorative art

- Masks and figures played an important role in prehistoric African art.
- Some were used in ceremonies and religious activities.
- Their meaning cannot be separated from their social or ceremonial purpose.
- Decorative works were also applied to textiles.
- Wooden objects could be decorated with beads, shells, clay, metal, feathers and ivory.

32. WHY DID PEOPLE MAKE ART?

- Art may have been made to beautify living spaces.
- It could express ideas, emotions or experiences.
- It may have provided amusement or occupied free time.
- Art could serve as a means of self-expression.
- Cole's discussion identifies deeper motives, including 'wishful thinking' and 'sympathetic imaging'.
- Wishful thinking: people may have represented what they wanted to happen, such as a successful hunt.
- Sympathetic imaging: people may have believed that representing a desired event could help bring it about.
- Artwork could commemorate important events such as battles, hunts or other major occurrences.
- Art can therefore be interpreted as having aesthetic, emotional, social, ritual and symbolic functions.

33. DATING AFRICAN ROCK ART

- Dating rock art is difficult because paintings and engravings are exposed to environmental processes.
- European climatic fluctuations provide some relative chronological clues because certain animals were associated with particular climatic periods.
- For example, paintings of woolly mammoths can be linked to periods before their extinction at the end of the Ice Age.
- African climatic conditions differed from those in Europe, making the same approach more difficult.
- The module notes that African climates were generally warmer, meaning people spent less time in caves than people in cold European environments.
- Because of these differences, dating African rock art requires careful interpretation of available archaeological and environmental evidence.

34. THREATS TO AFRICAN PREHISTORIC ART

- Many African art forms and archaeological representations are under threat.
- Population growth can expose sites and materials to damage.
- Vandalism is a major threat to cultural heritage.
- Protection and conservation are necessary to preserve archaeological and artistic evidence for future generations.

35. QUICK EXAM REVISION POINTS

- Fossil + context = better interpretation of age, environment and association.
- Taphonomy = processes affecting an organism from death to discovery.
- Stratigraphy = study of layers and their sequence.
- Superposition = lower undisturbed layers are older than overlying layers.
- Cross-cutting = the cutting feature is younger than the feature cut.
- Faunal succession = recognizable sequences of fossils through strata.
- GTS = chronological framework of Earth history using eons, eras, periods and epochs.
- Bipedalism = habitual locomotion on two legs.
- Major bipedalism theories = locomotor efficiency, thermoregulation, freeing hands and postural feeding.
- Oldowan = Mode 1; simple flaking and core tools.

- Acheulian = Mode 2; bifacial handaxes, picks and cleavers.
- MSA = strongly associated with Mode 3 prepared-core technologies and increasing regional variation.
- LSA = Mode 4 emphasis on blades, microliths and increased bone/antler/ivory/shell use.
- Out of Africa = modern humans originated in Africa and subsequently dispersed.
- Multiregional continuity = regional populations contributed to the evolution of modern humans after Homo erectus dispersal.
- Behavioral modernity = increased technological diversity, symbolism, art, personal adornment, specialized tools and complex social behavior.
- African prehistoric art = painting, engraving, pottery, molding, masks, figures and adornment.
- Major motives for art in the module = expression, beautification, amusement, wishful thinking, sympathetic imaging and commemoration.

SELF-TEST QUESTIONS FROM THE MODULE

Modern human origins

- How does the Multiregional Continuity Model explain the origin of modern humans?
- Explain the Out of Africa model and its claim of a recent African origin for humanity.
- How is anatomical evidence used to explain the origin of modernity in humans?
- What were the technological similarities and differences between MSA Homo sapiens in Africa and Middle Paleolithic Neanderthals?
- Discuss the evidence and debate concerning possible interbreeding between Neanderthals and modern humans.

Stone tools

- How does the MSA differ from the ESA?
- Why is it difficult to determine who made particular tools during some periods?
- What are the defining differences between the MSA and LSA?
- How do technological changes in stone-tool production indicate changes in cognition?

Prehistoric art

- What were the major motivating factors of prehistoric art makers?
- Why does continental Africa possess less cave art than Europe according to the module?
- Discuss the evidence used to argue that African art is very old.
- Why is prehistoric art in Africa under threat?

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Source note: These revision notes were prepared from the uploaded AHT 212 module. Where the module presents debated dates, interpretations or terminology, the notes retain that framing rather than silently substituting outside information.